

FACTSHEET

Cancer Incidence and Mortality in Mayurbhanj district (2023), Odisha state, India



Tata Memorial Centre (TMC), Mumbai
(A Grant-in-Aid Institution under Department of Atomic Energy, Government of India)
Department of Health and Family Welfare, Government of Odisha
Pandit Raghuram Murmu Medical College & Hospital, Baripada
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FEATURES OF MAYURBHANJ DISTRICT

- » Mayurbhanj district is the largest district and the third most populous of Odisha state, India.
- » About 60% of population are Scheduled Tribes.
- » Two in three males and one in three females aged 15 years or above use tobacco.
- » Almost half of all males and one in six females aged 15 years or above use alcohol.
- » Mayurbhanj district is administratively divided into 26 blocks, with 3944 villages and 70 urban wards. More than **90% of the district lives in rural areas.**



OVERVIEW

- » The Population-based Cancer Registry (PBCR) for Mayurbhanj district was established in 2023 by Tata Memorial Centre, Mumbai, with the support of National Institute for Science, Education and Research, Bhubaneswar, Department of Health and Family Welfare, Government of Odisha, and Pandit Raghuram Murmu Medical College and Hospital, Baripada.

OBJECTIVE

- » Mayurbhanj PBCR aims to systematically collect and present data on cancer incidence and mortality in the district, for usage in cancer control policy and programme planning.

METHODS

- » Government of Odisha issued administrative orders to ensuring cooperation to data collection process from public health system and also from private hospitals.
- » Confirmed cases are documented in a prescribed format and entered in CanReg5 software, developed by International Agency for Research on Cancer (IARC), Lyon, France.
- » The primary site, basis of diagnosis, and histology are coded based on the International Classification of Diseases for Oncology – Third Edition (ICD-O-3).



SOURCES FOR DATA

- » Major hospitals in Khordha, Cuttack, Keonjhar, Balsore, Jharkhand, and TMH Mumbai.
- » Clinics, Labs and Imaging centres within and outside Mayurbhanj district.
- » Vital Statistics office
- » Community interaction with the help of health workers like Auxiliary Nurse Midwives (ANMs) and Accredited Social Health Activists (ASHAs).

Estimated population

27.78 Lakhs

Male: 13.79 Lakhs

Females: 13.99 Lakhs

FINDINGS OF THE YEAR 2023

Incidence cases

1,170

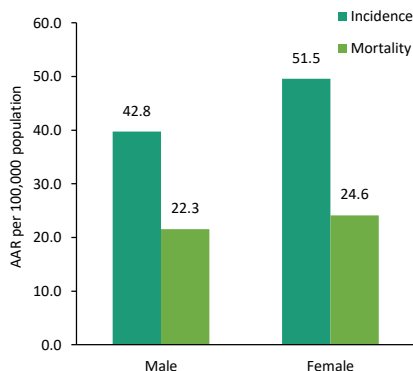
(519 males, 651 females)

Mortality cases

571

(265 males, 306 females)

Age adjusted incidence and mortality rates per 100,000



One in 21 males are at risk of developing cancer before the age of 75 years

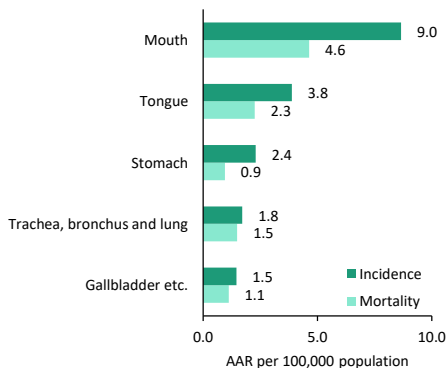


One in 18 females are at risk of developing cancer before the age of 75 years

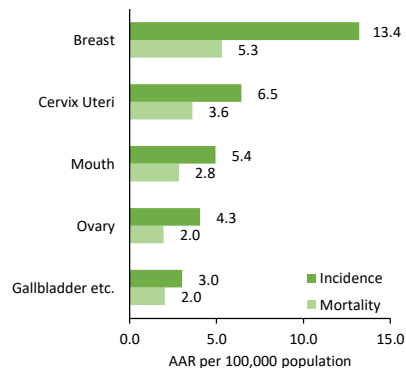


Leading sites of cancer incidence among

Males

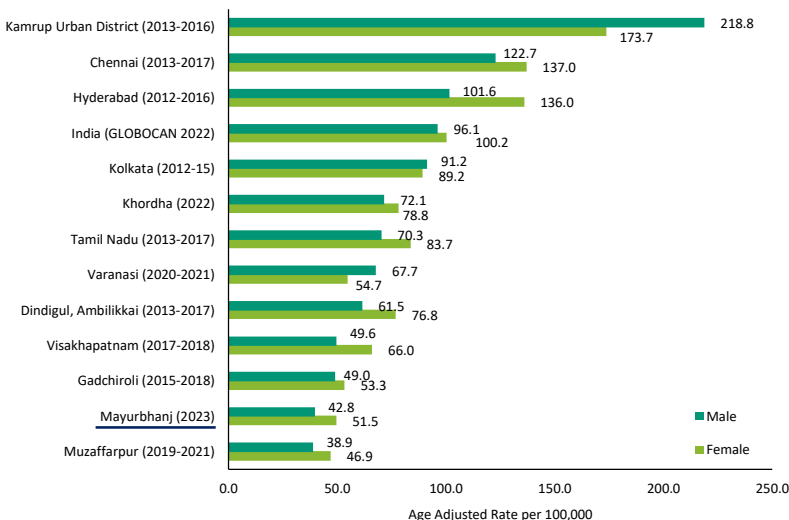


Females



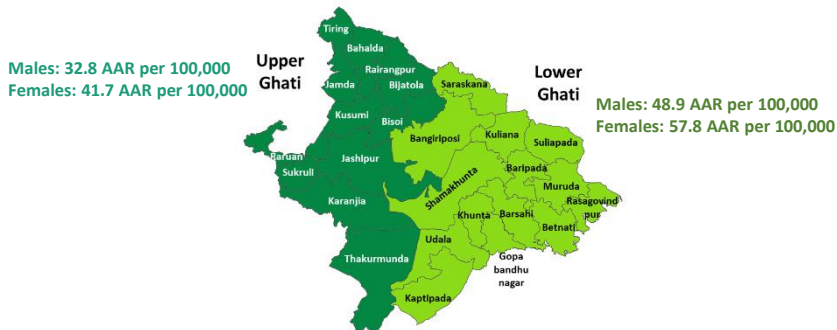
- » **Tobacco-related cancer cases:** 209 (40.3%) of all cancers in males (two in five cases) and 94 (14.4%) of all cancers in females (one in 7 cases).
- » **Paediatric cancer incident cases:** 21 (1.8%) cases occurred in the age group 0-14 years, with an Age Adjusted Incidence rate of 33.7 per million and 12.9 per million for boys and girls, respectively.

COMPARISON WITH OTHER INDIAN REGISTRIES



- » Cancer Incidence in Mayurbhanj district is much lower than that of Khordha district in Odisha, but is comparable to Muzaffarpur, rural area in Bihar and Gadchiroli, a tribal district in Maharashtra.

COMPARISON WITHIN MAYURBHANJ DISTRICT



- » About 70% of both incidence and mortality cases of Mayurbhanj district belong to the Lower Ghati zone, due to its larger population. However, this difference persists between the incidence AAR of these zones.
- » Upper Ghati reports 28-33% lower incidence than the Lower Ghati (males: RR 0.67, 95% CI 0.56-0.80; females: RR 0.72, 95% CI 0.62-0.85).
- » This can be attributed to the unavailability and inaccessibility of cancer diagnostic and treatment centres in interior regions of Mayurbhanj district, reflecting under-diagnosis or delayed reporting of cancer cases, and a difference in environmental risk factors.

INSIGHTS AND RECOMMENDATIONS

- » Strengthening of registry operations with the support of the district administration and state government shall improve the registry's process and reduce underreporting in the coming years.
- » To address underdiagnosis of cancer cases in the district, improvements in health infrastructure and strengthening of early detection programme play a crucial role. Moreover, **patient navigation** efforts are highly effective in helping patients approach cancer treatment and improving treatment compliance. **It is recommended to strengthen patient navigation process for the benefit of public health.**
- » The high incidence of **oral cancers** can be correlated to the high prevalence of **tobacco and alcohol** use among males of Mayurbhanj district. Oral cancer is completely preventable; primary prevention by strengthening tobacco and alcohol cessation, secondary prevention by strengthening early diagnosis are highly recommended to reduce the incidence and improve the survival of oral cancer in the coming years.
- » As **gastrointestinal cancers** like stomach and gall bladder make up a significant proportion of the cancer burden in Mayurbhanj district, it is recommended to raise awareness about the common risk factors as well as signs and symptoms of these cancers to reduce time to diagnosis. Additionally, **improving health infrastructure and resources for diagnosis and treatment of gastrointestinal cancers is highly recommended.**
- » Considering the high burden of oral, breast, and cervical cancers, Tata Memorial Centre initiated an **Early Detection Programme (EDP)** in Mayurbhanj district. In the period August 2023 to May 2025, 5,768 individuals have been screened. Individuals with screen-positive findings have been referred to higher cancer centres and are under regular follow-up. 72 cancer cases (51 oral, 20 breast, and one cervical cancer) have been diagnosed by EDP in this period, mostly in locoregional stage. Continuation and strengthening of EDP is highly recommended not only to complement the PBCR, but also to support the population of Mayurbhanj in accessing the right care and the right time.

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